CivisMetric V2 ENTERPRISE SUITE

Next-Generation Civil Engineering Suite, CAD Dashboard & Estimation Engine

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01 Executive Overview

CivisMetric V2 is a unified, full-stack digital engineering platform engineered to eliminate calculation errors, streamline material quantification, automate Bar Bending Schedules (BBS), and accelerate multi-disciplinary construction project management.

By integrating modern web technologies (**React 19**, **Vite**, **Three.js**, and **Express.js**) with rigorous civil engineering standards, CivisMetric V2 empowers structural engineers, quantity surveyors, site managers, and contractors with 33+ computational modules and a live interactive 2D/3D CAD studio.

📐 33+ Specialized Calculators

Complete mathematical coverage for core materials, structural concrete, steel reinforcement, finishing, and MEP sizing.

⚡ 2D/3D WebGL CAD Studio

Real-time drafting canvas with multi-layering, snap-to-grid, vector JSON export, and 3D wireframe inspection.

📊 Auto-Aggregated Master BOQ

Continuous synchronization of quantities and project schedules with live Schedule of Rates (SOR) pricing.

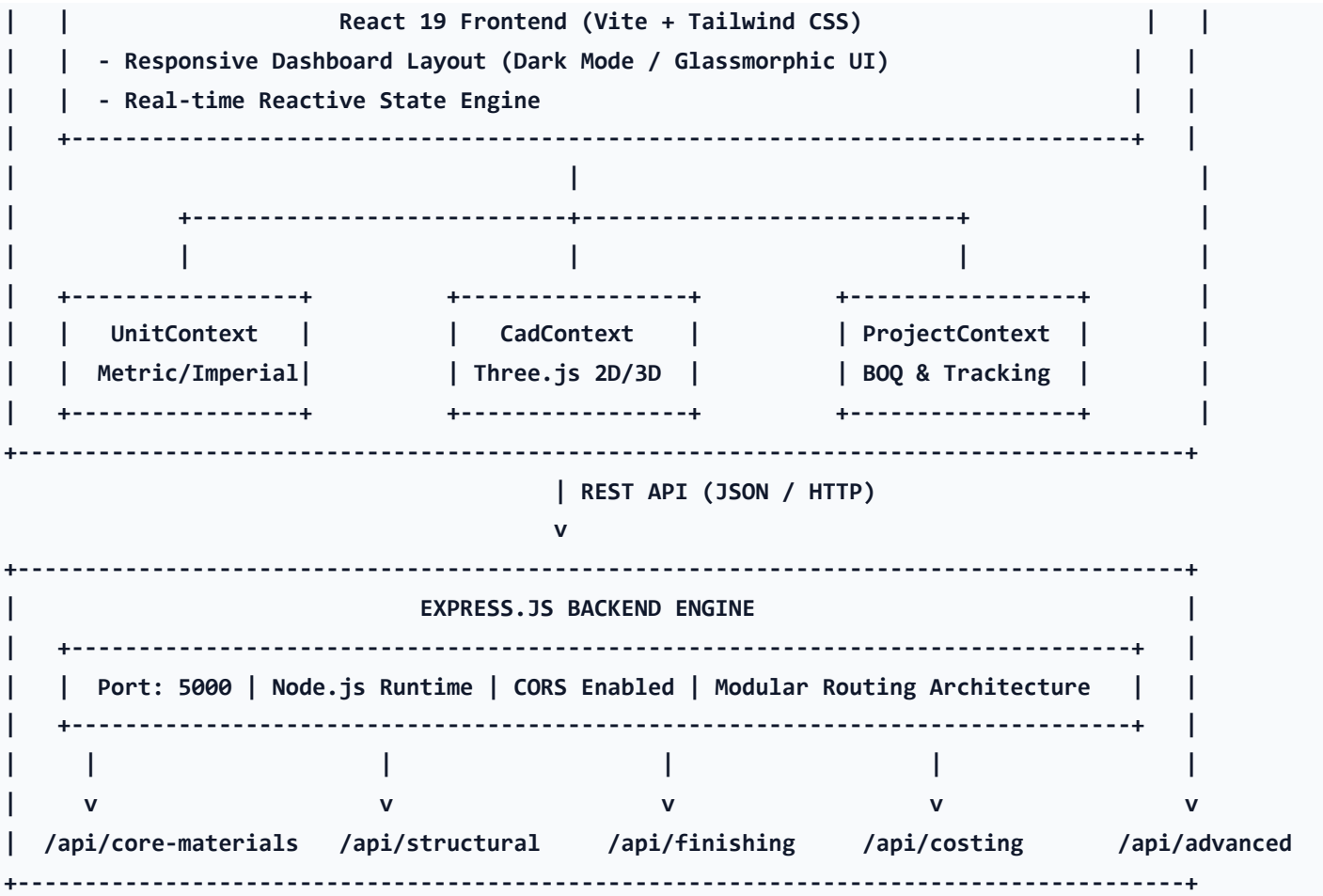
✂️ Rebar Waste Minimization

1D Cutting Stock Optimization algorithm reducing commercial 12m steel rebar scrap from 15% to under 2.5%.

02 System Architecture & Tech Stack

CivisMetric V2 adopts a decoupled client-server micro-route architecture designed for rapid responsiveness, zero-lag reactive input processing, and modular scalability.





COMPONENT	TECHNOLOGY	VERSION	ROLE & RESPONSIBILITY
Frontend Core	React.js	`^19.2.4`	Component hierarchy, state orchestration, fast re-renders.
Build Pipeline	Vite	`^8.0.1`	Lightning-fast ESM development server and optimized bundle generator.
CAD Engine	Three.js	`^0.183.2`	WebGL rendering, geometric scene graph, 2D/3D vector manipulation.
Design System	Tailwind CSS	`^3.4.1`	Glassmorphism, custom engineering dark theme, micro-animations.
Backend API	Express.js / Node	`v20.x`	Modular REST micro-routes, material cost evaluations, BBS logic.

03 Engineering Modules & Features

1. CAD & Project Master

- **AutoCAD Studio (`CadStudio.jsx`):** Real-time 2D/3D drawing workspace featuring Point, Line, Polyline, Rectangle, Circle, Arc, Dimension, and Annotation tools with coordinate grid snapping and multi-layer management.
- **Project Tracker (`ProjectTracker.jsx`):** Milestone schedule management with Gantt charts, budget variance tracking, and task completion percentage.
- **Project Master BOQ (`ProjectMasterBOQ.jsx`):** Centralized bill of quantities compiling excavation, concrete, steel, masonry, and finishing estimates with one-click export.
- **BBS Rebar Optimizer (`RebarCuttingOptimizer.jsx`):** Dynamic linear cutting stock algorithm solving 1D bin-packing on 12-meter commercial stock lengths to slash off-cut scrap.
- **IS 10262 Mix Design (`ConcreteMixDesignIS10262.jsx`):** Implementation of Indian Standard IS 10262:2019 mix proportioning for M15 to M60 grades.

2. Core Construction Materials

MODULE	CORE FORMULA / ALGORITHM	OUTPUT PARAMETERS
Cement Calculator	$$\text{Dry Vol} = \text{Wet Vol} \times 1.54$$ $$\text{Bags} = \lceil (\text{Ratio} / \sum \text{Parts}) \times \text{Dry Vol} \times 1440 / 50 \rceil$$	Bag count (50kg), total dry mass (kg), cost.
Sand Calculator	$$\text{Adjusted Vol} = \text{Vol} \times (1 + \text{Bulking \%}/100)$$ $$\text{Tons} = (\text{Adjusted Vol} \times 1550) / 1000$$	Bulked volume (m^3), dry weight (MT), Brass (100 cft).
Aggregate Calculator	$$\text{Gross Vol} = \text{Vol} \times \text{Compaction Factor (1.20)}$$ $$\text{Weight} = (\text{Gross Vol} \times 1600) / 1000$$	Order volume (m^3), Tonnage (MT), truckloads.
Bricks Calculator	$$\text{Bricks} = \frac{\text{Net Wall Vol}} {\text{Nominal Brick Vol}} \times (1 + \text{Wastage \%})$$	Brick units count, dry mortar volume, cement & sand split.
Steel Calculator	$$W = \frac{D^2}{162.28}\text{ kg/m} \quad (\text{Imperial: } d^2/52.16\text{ lbs/ft})$$	Unit weight (kg/m), total weight (kg & MT), total cost.

3. Structural Elements Quantification

- **Foundation (`FoundationCalculator.jsx`):** Concrete volume, excavation boundary margins, base PCC leveling bed, and bottom mesh reinforcement.
- **Excavation (`ExcavationCalculator.jsx`):** Pit and trench excavations incorporating soil angle of repose (slope batter), swell factor, and backfill volume.
- **Column (`ColumnCalculator.jsx`):** Rectangular and circular columns, longitudinal vertical bars, lateral tie/spiral pitch, and shuttering formwork area.
- **Slab (`SlabCalculator.jsx`):** One-way and two-way structural slabs, main and distribution reinforcement, concrete wet volume.
- **Beam (`BeamCalculator.jsx`):** Clear span flexural members, top compression bars, bottom tension bars, 2-legged shear stirrups.
- **Staircase (`StaircaseCalculator.jsx`):** Waist slab inclination length, riser/tread volume, landing slabs, concrete and steel detailing.

04 Backend REST API Specification

All endpoints are mounted on `http://localhost:5000` and accept standard `application/json` payloads.

GET **/api/health** — Server Health & Uptime Status

POST **/api/core-materials/concrete** — Grade Mix Breakdown

```
// Request Body:
{
  "wetVolumeM3": 25.0,
  "mixRatio": [1, 1.5, 3],
  "pricePerM3": 4500
}

// Response Body:
{
  "success": true,
  "wetVolumeM3": 25.0,
  "dryVol": "38.50",
  "cementBags": 202,
  "sandTons": "16.80",
  "aggTons": "33.60",
  "cost": "112500.00"
}
```

POST **/api/structural/column** — Column Volume & Formwork

```
// Request Body:
{
  "length": 0.45,
  "width": 0.45,
  "height": 3.6,
  "isCircular": false,
  "count": 12,
  "costPerCubicMeter": 4800
}
```

```
// Response Body:
{
  "success": true,
  "volume": "8.75",
  "cost": "41990.40"
}
```

POST **/api/advanced/rebar-bbs** — Rebar Weight & BBS Sizing

```
// Request Body:
{
  "diaMm": 20,
  "cuttingLenM": 6.5,
  "numBars": 40,
  "ratePerKg": 70
}

// Response Body:
{
  "success": true,
  "unitWt": "2.469",
  "totalWeightKg": "641.98",
  "cost": "44938.27"
}
```

05 State Management Architecture

CivisMetric V2 implements reactive, decoupled context providers ensuring that unit conversions, CAD vector operations, and project metadata sync seamlessly without re-render bottlenecks.

UnitContext (Dual Units Matrix)

Instant toggle between Metric (SI) and Imperial. Automatically recalculates all input boundaries, conversions, and unit labels in real-time.

CadContext (Vector & Scene Graph)

Maintains Three.js 2D/3D entities, layers, active drafting tool, grid snap step, pan/zoom transformation, and undo/redo stacks.

ProjectContext (Persistent State)

Houses client details, milestone timelines, continuous BOQ aggregation, and synchronizes with browser LocalStorage.

06 Installation, Setup & Deployment Guide

1. Backend Setup

```
cd "d:\civis_metric  V2\engineering-backend"
npm install
node server.js
# API Server active on http://localhost:5000
```

2. Frontend Setup

```
cd "d:\civis_metric  V2\my-engineering-dashboard"
npm install
npm run dev
# Dashboard UI active on http://localhost:5173
```

3. Production Bundle Build

```
npm run build
# Creates optimized production assets in dist/ directory
```

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Compliant with IS 456:2000, IS 10262:2019, IS 1200, SP 34, BS 8666, and ACI 318.